

Cyclistic Bike-Share Case Study

Understanding Casual Rider Behavior to Increase Annual Memberships

3.97M valid trips analyzed | Aug 2025–Jul 2026 | SQL • MySQL • Excel •
Tableau

Business Objective

Identify behavioral differences between casual riders and annual members and use those insights to identify opportunities for membership conversion.

Key Findings

- 34.9% of trips were made by casual riders.
- Weekend and warmer-month usage has a substantially higher casual share.
- Longer-duration trips have a substantially higher casual share.
- 42 high-volume stations meet the selected criteria for potential targeted campaigns.

Recommended Strategy

Target casual riders when and where they are most active—particularly during warmer months, weekends, and at high-volume stations with a high casual-trip proportion—and test membership messaging focused on the value of converting recurring casual usage into annual membership.

Ask

Business Task:

Increase the number of annual memberships at Cyclistic by identifying opportunities to convert casual riders into annual members.

Key Business Questions

The analysis will answer the following questions:

1. How does overall trip activity differ between casual and member riders?
2. When do casual and members riders use Cyclistic bikes most frequently?
3. How do they differ in trip duration?
4. Which stations have sufficient trip volume and a meaningful proportion of casual riders to be considered potential locations for targeted membership-conversion campaigns?
5. Where are these potential target stations geographically concentrated?

Key Stakeholders:

- Lily Moreno — Director of Marketing
- Cyclistic Marketing Analytics Team
- Cyclistic Executive Team

Prepare

Data source

The data came from Divvy which is the bicycle sharing system in the Chicago metropolitan area, it is publicly available online and stored in an Amazon Web Services as compressed .zip files ([Download](#)) and contain Cyclistic historical trip data.

Data selected

The selected dataset contains Cyclistic trip data for a 12-month period, from **August 1, 2025, through July 31, 2026**, representing the most recent complete 12-month period available at the time of analysis. Twelve monthly files were downloaded and decompressed. The files follow the naming convention **YYYYMM-divvy-tripdata.zip**. The following table presents the name and trip records count of each monthly file, along with the **total number of trip records** across the selected months.

Seq.	File Name	Records count
1	202607-divvy-tripdata	869,051
2	202606-divvy-tripdata	762,550
3	202605-divvy-tripdata	653,704
4	202604-divvy-tripdata	484,252
5	202603-divvy-tripdata	317,037
6	202602-divvy-tripdata	201,450
7	202501-divvy-tripdata *	137,787
8	202512-divvy-tripdata	140,534
9	202511-divvy-tripdata	356,628
10	202510-divvy-tripdata	646,039
11	202509-divvy-tripdata	714,759
12	202508-divvy-tripdata	790,177
Total records count:		6,073,968

Data organization

The data CSV file contains trip records and each record has the following fields:

Seq.	Field name	meaning
1	ride_id	Unique identifier for a trip
2	rideable_type	Bike type
3	started_at	Ride start date/time
4	ended_at	Ride end date/time
5	start_station_name	Starting station name
6	start_station_id	Starting station ID
7	end_station_name	Ending station name
8	end_station_id	Ending station ID
9	start_lat	Starting Latitude (Decimal Degrees)
10	start_lng	Ending longitude (Decimal Degrees)
11	end_lat	Starting Latitude (Decimal Degrees)
12	end_lng	Ending longitude (Decimal Degrees)
13	member_casual	Rider type

Data credibility

The data comes from Cyclistic/Divvy's official historical trip-data repository, making it a credible primary source for analyzing actual bike-share usage and comparing the riding behavior of casual and member riders.

Potential Data Bias and Quality Issues

Potential sources of bias and data-quality issues will be investigated during the Process phase before drawing conclusions. The following checks will be performed:

The data should include trips from a broad range of stations and locations rather than being concentrated in only a few stations or areas.

- The data should cover all days of the week, including both weekdays and weekends, so that differences in riding behavior can be properly analyzed.
- There should be sufficient trip records for both casual riders and annual members to allow meaningful comparisons between the two groups.
- Invalid ride durations: Identify zero, negative, or unusually long ride durations and investigate whether they represent valid trips or data-quality issues.

Data ROCCC

- The data is considered as **Reliable** because it is published by Divvy bicycle sharing system for public use.
- The data is provided by Divvy and not collected by third party, so it is **Original**.
- It contains the needed fields for bikes trips, but it doesn't contain any information about customer, and since the task is to understand how casual riders and annual members use Cyclistic bikes differently we can say that data is mostly **Comprehensive** to get the differences in bikes use.
- Dataset is **Current** and contains the last 12 months trips.
- The data is **Cited** because it is sourced and provided by Divvy/Cyclistic and located at AWS repository. Also The data has been made publicly available by Motivate International Inc.

Privacy

The published trip data is **anonymized** and does not contain personally identifiable information such as names or contact details. The data will not be combined with other sources for the purpose of identifying individual riders.

Security

The data was downloaded from the official source and stored securely. Since the dataset is publicly available and anonymized, the privacy risk is limited; however, standard data-security practices will be followed to protect the downloaded files and analysis.

Licensing

The data is provided under the **Bikeshare Data License Agreement**, which grants users a non-exclusive, royalty-free, limited, perpetual license to access, reproduce, analyze, copy, modify, distribute, and use the data for lawful purposes, subject to the terms of the agreement. The analysis will comply with the applicable data-use and attribution requirements.

[Bikeshare Data License Agreement](#)

Accessibility

The final analysis and visualizations will use clear titles and labels, readable text, appropriate color contrast, and straightforward language so that the findings can be understood by both technical and non-technical stakeholders.

Data integrity

Data integrity checks will be performed during the Process phase before conducting the analysis and drawing conclusions. The following checks will be performed:

- Missing values: Identify missing values, particularly in key fields such as rider type, ride start/end times, and station information.
- Duplicate records: Check for duplicate trip records.
- Invalid dates and times: Check for missing, incorrectly formatted, or logically invalid timestamps, such as an end time occurring before the start time.
- Inconsistent categories: Verify that categorical variables, such as rider type and bike type, contain consistent and expected values across all datasets.

Process

The Process phase focused on **cleaning, validating, transforming, and preparing the Cyclistic trip data for analysis**. The objective was to improve data quality and ensure that the dataset was sufficiently accurate, consistent, and complete for comparing the riding behavior of casual riders and annual members.

The dataset covered the period **August 2015 through July 2016** and initially contained **6,037,968 trip records**.

The data preparation and cleaning process consisted of four sub-steps:

Initial Data Quality Checks

At this stage, the original data was examined for potential quality issues, including **missing values, duplicate records, date and time formatting, date-range consistency, category consistency, and geographic coordinate validity.**

Initial Data Quality Checks		
Initial number of trip records: [6,073,968]		
Check point	Result	Action
Missing values	Missing start station id: 1273200 trips. Missing end station id: 752047 trips.	deleted.
Duplicate records	Duplicate records identified: 20 trips	deleted
date and time formatting	All date/time is in valid syntax.	No Action
Invalid duration	Rides with start time after end time: 19	deleted
Date range	Min date: 2025-08-01 Max date:2026-07-31 Consistent with the monthly dataset.	No action
Bike type	Two bike types are present: classic and electric.	No action
Rider type	Two rider types are present: member and casual.	No action
geographic coordinate validity	All records detected in the first step with Missing start or end station id had only one or two decimal digits in start longitude, so the precision of the physical distance isn't less than 830m in Chicago city which considered as invalid values.	No action (they were deleted in the first step)
File naming	The file named 202501-divvy-tripdata contains trip data for January 2026 rather than January 2025.	Correct the file name.
resulted dataset: After cleaning, ' 4,012,682 ' trip records remained for analysis.		

Data Transformation and Derived Variables

Several additional columns were created to support data cleaning and analysis. A unique row_id was added to each record to provide a reliable row-level identifier and facilitate duplicate identification and removal. A day_of_week column was derived from the started_at timestamp to support temporal analysis. ride_duration_seconds was calculated from the difference between ended_at and started_at.

These derived variables enabled additional data-quality checks that could not be performed directly using the original variables. The following table shows the added new columns and purpose for each one.

New column	Purpose	Method
row_id	Row-level identification	Auto-incrementing ID
day_of_week	Day-of-week analysis	Derived from started_at
ride_duration_seconds	Ride duration	ended_at - started_at
ride_speed_kmh	Average ride speed	Distance ÷ duration

Derived-Variable Quality Checks

After calculating the derived variables, additional checks were conducted to identify potentially invalid or anomalous rides. Records with zero durations were removed because they indicate invalid or inconsistent start and end timestamps. Rides with a duration of less than one minute were treated as potentially anomalous and excluded from the analysis based on the defined cleaning criterion.

In addition, rides with calculated average speeds above 35 km/h were treated as implausible outliers for this bike-share analysis and removed. The 35 km/h threshold was used as a data-cleaning criterion rather than an absolute physical limit for bicycles. The maximum distance of the trip was also calculated and considered a reasonable value compared to its duration.

Derived-Variable Quality Checks		
Initial number of trip records: 4,012,682		
Check point	results	action
Ride Duration	There are 173 trips with zero-durations	deleted
	There are 43096 trips with less the one minutes duration.	deleted
Ride distance	Max tripe distance: 48,370 Km. Trips with distances greater than 5 km were reviewed against their recorded trip durations. No obvious distance-duration inconsistencies were identified.	No action
resulted dataset	3,969,413 trip records remain after cleaning.	Dataset used for analysis

Bias and Coverage Checks

The dataset was also examined for potential sources of bias and coverage limitations. These checks included reporting-period coverage, weekday/weekend distribution, geographic coverage, rider-type distribution, and bike-type distribution.

The following tables summarize the checks performed, results identified, and actions taken during the data preparation and cleaning process.

Bias and Coverage Checks		
Initial number of trip records: 3,969,413		
Check point	results	action
Weekday/Weekend Coverage	Trips cover all seven days of the week, including weekdays and weekends.	No action
Geographic Coverage	There are 1932 different start stations. There are 1983 different end stations. So, data includes a broad station coverage.	No action
Rider Type Distribution	The casual rider percentage: 34.8% The member rider percentage: 65.2% Both groups have sufficient records for comparison.	No action
Final dataset	3,969,413 trip records remain after cleaning.	Dataset used for analysis

After completing the data-quality and validation checks, **3,969,413** trip records remained in the final dataset for analysis. The cleaned dataset contains the required information to compare casual and member riding behavior.

Analysis

In this phase, the cleaned and validated Cyclistic trip data is analyzed to understand how casual riders differ from annual members in their usage behavior. The analysis focuses on when, where, and how riders use Cyclistic bikes, including patterns by month, day of week, hour, trip duration, bike type, and stations.

The objective is not only to identify statistical differences, but also to determine which differences are relevant to Cyclistic's goal of increasing annual memberships. These insights will form the evidence base for the recommendations presented in the Share and Act phases.

Rider Type Distribution

The analysis begins by examining the overall distribution of trips between casual riders and annual members. This provides a baseline for understanding the relative contribution of each rider group to Cyclistic's total trip volume. Establishing this baseline also provides context for the subsequent analyses, which investigate how the usage patterns of the two rider groups differ across time, trip characteristics, bike type, and location. The following table shows the number and percentage of total trips made by each rider type during the 12-month study period.

Trip Count and Percentage by Rider Type		
rider type	Trip count	Percentage of total trips
Casual	1384377	34.88%
Member	2585036	65.12%

The table shows that member trips are approximately twice as numerous as casual trips.

Monthly Trip Distribution by Rider Type (%)												
Year	2025					2026						
Month	8	9	10	11	12	1	2	3	4	5	6	7
Casual %	42.9	36.8	34.33	27.3	19.6	17.05	20.09	27.31	28.94	36.39	39.13	40.18
Member%	57.1	63.9	65.67	72.7	80.4	82.95	79.91	72.69	71.06	63.61	60.87	59.82

The Table shows a strong seasonal pattern in the composition of trips, Casual trips are highest in warmer months and decline substantially through the colder ones. Conversely, the member share follows the opposite pattern.

Daily Distribution of Trips by Rider Type		
Day of week	Casual (%)	Member (%)
Monday	30.1	69.9
Tuesday	26.9	73.1
Wednesday	27.2	72.8
Thursday	29.18	70.82
Friday	36.37	63.63
Saturday	46.1	53.9
Sunday	48.42	51.58

There is a clear difference between weekdays and weekends. the casual share increases substantially toward the weekend. while Members show the opposite pattern.

Hourly Distribution of Trips by Rider Type			
Hour of day	Trip Count	Casual (%)	Member (%)
0:00	45650	56.53	43.47
1:00	27905	59.00	41.00
2:00	16794	61.73	38.27
3:00	9791	59.33	40.67
4:00	10190	45.61	54.39
5:00	34537	24.79	75.21
6:00	98774	19.44	80.56
7:00	195785	18.13	81.87
8:00	246149	20.42	79.58
9:00	167681	29.38	70.62
10:00	163029	37.50	62.50
11:00	195999	39.75	60.25
12:00	227702	40.16	59.84
13:00	229037	41.25	58.75
14:00	234673	42.08	57.92
15:00	282707	38.25	61.75
16:00	375861	33.37	66.63
17:00	415011	31.82	68.18
18:00	323430	33.85	66.15
19:00	224011	35.40	64.60
20:00	158788	36.40	63.60
21:00	125437	39.15	60.85
22:00	96644	43.96	56.04
23:00	63828	48.66	51.34

The table shows that Casual riders account for the majority of trips between midnight and 04:00, while members dominate most other hours.

As a result, the Rider Type Distribution founded that:

- Casual trips are approximately half as numerous as member trips.
- Casual riders represent a much larger share of trips during warmer months and a much smaller share during winter.
- Casual riders have a substantially higher share of trips on weekend, while members dominate weekday trips.
- Casual riders account for the majority of trips between midnight and 04:00, while members dominate most other hours.

Trip Duration Distribution by Rider Type

Trip duration is analyzed to identify differences in how long casual riders and members use Cyclistic bikes. Comparing the distribution of trip

durations between the two rider types can help determine whether one group tends to take shorter or longer trips and identify duration ranges where their usage patterns differ significantly.

The following table shows the number and percentage of casual and member trips across different trip-duration ranges during the 12-month study period.

Trip Duration Distribution by Rider Type					
Duration	casual	member	total	casual(%)	member(%)
<30 M	1147053	2431311	3578364	32.1	67.9
30-45 M	110188	112512	222700	49.5	50.5
45M-1H	49840	24139	73979	67.4	32.6
1-2H	59999	11838	71837	83.5	16.5
2-4H	13647	2890	16537	82.5	17.5
> 4H	3650	2346	5996	60.9	39.1

The table shows that casual riders account for a particularly high share of trips lasting more than 45 minutes, while members account for a higher share of shorter-duration trips.

Bike Type Usage by Rider Type

This analysis examines how casual riders and members use Cyclistic's different two bike types (electric/classic). Comparing bike-type usage between the two rider groups helps identify whether their preferences differ and whether a particular bike type is more commonly used by one group than the other.

Bike Type Usage by Rider Type					
	Casual	Member	Total	Casual (%)	Member (%)
Electric Bike	776525	1307190	2083715	37.27	62.73
Classic Bike	607852	1277846	1885698	32.23	67.77

The table shows that electric bikes account for approximately 5.4 percentage points more of casual trips than member trips.

Interactive Station-Level Analysis

To identify potential stations for targeted membership-conversion campaigns, an interactive **Tableau dashboard** was created to analyze station-level trip activity. The dashboard displays station ID, names, locations and provides filters for total trip volume and casual-trip proportion.

For the target-station analysis, the dashboard was filtered to stations with **at least 5,000 total trips** during the 12-month study period and a **minimum casual-trip proportion of 40%**. These criteria were used to

identify stations with sufficient trip volume where **casual riders represent a meaningful share of usage** and may therefore provide opportunities for targeted membership-conversion campaigns. The criteria showed that there are forty two potential target stations concentrated primarily in **central Chicago and areas along or near the Lake Michigan lakefront**, including downtown, South Loop, River North, Lincoln Park, and Lakeview. This geographic concentration provides a practical starting point for targeted membership-conversion campaigns.

A full list of these stations, including their relevant information, is provided in **Appendix A**.

Interactive Tableau Dashboard: [Explore the Station-Level Analysis on Tableau Public](#)

Summary of Key Analysis Findings:

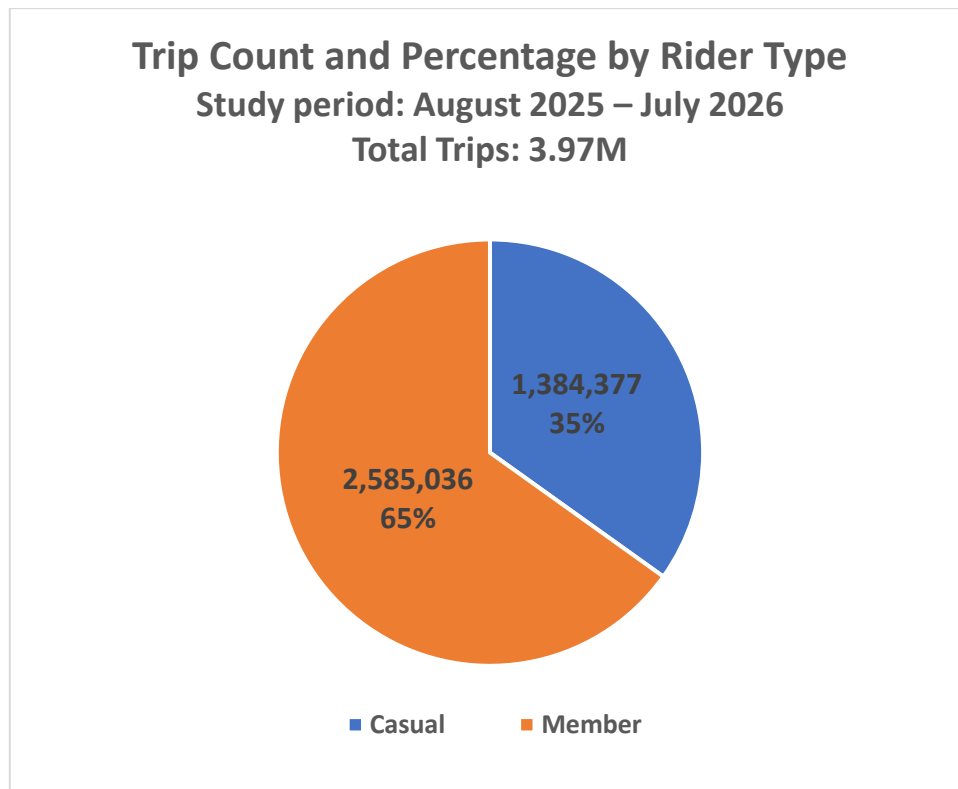
The analysis identified several clear differences in how casual riders and members use Cyclistic bikes:

1. **Rider composition:** Members account for approximately 65.1% of all trips, while casual riders account for 34.9%. Although members generate more trips overall, the substantial volume of casual trips represents an important potential audience for membership conversion.
2. **Seasonal and weekly patterns:** Casual riders account for a larger proportion of trips during the warmer months and on weekends, while members have a stronger presence during weekdays. This indicates that casual riders and members have different patterns of usage throughout the year and week.
3. **Time-of-day patterns:** casual riders have a particularly high share of trips during the first four hours of the day, while Members have a higher share of trips during most of the remaining hours.
4. **Trip duration:** Casual riders account for a particularly high proportion of longer-duration trips, while members have a higher proportion of shorter-duration trips.
5. **Bike type:** Electric bikes account for a slightly larger proportion of casual trips than classic bike trips (5.5%).
6. **Targeted stations:** 42 high-volume stations have at least 5,000 trips during the study period and a casual-trip proportion of at least 40%,

making them potential locations for targeted membership-conversion campaigns. These stations are concentrated primarily in central Chicago and areas along or near the Lake Michigan lakefront, including downtown, South Loop, River North, Lincoln Park, and Lakeview.

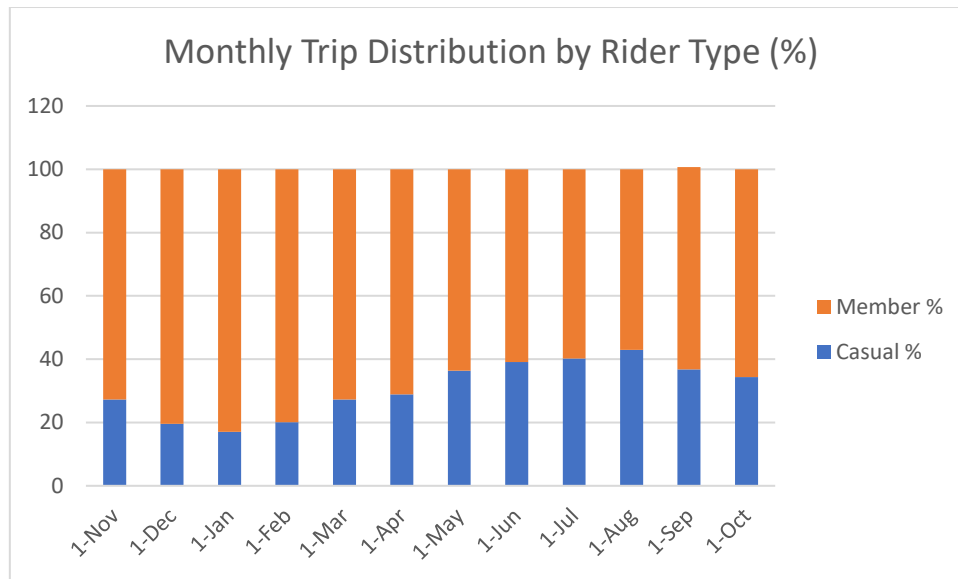
Share Phase

1. Who are Cyclistic's riders?

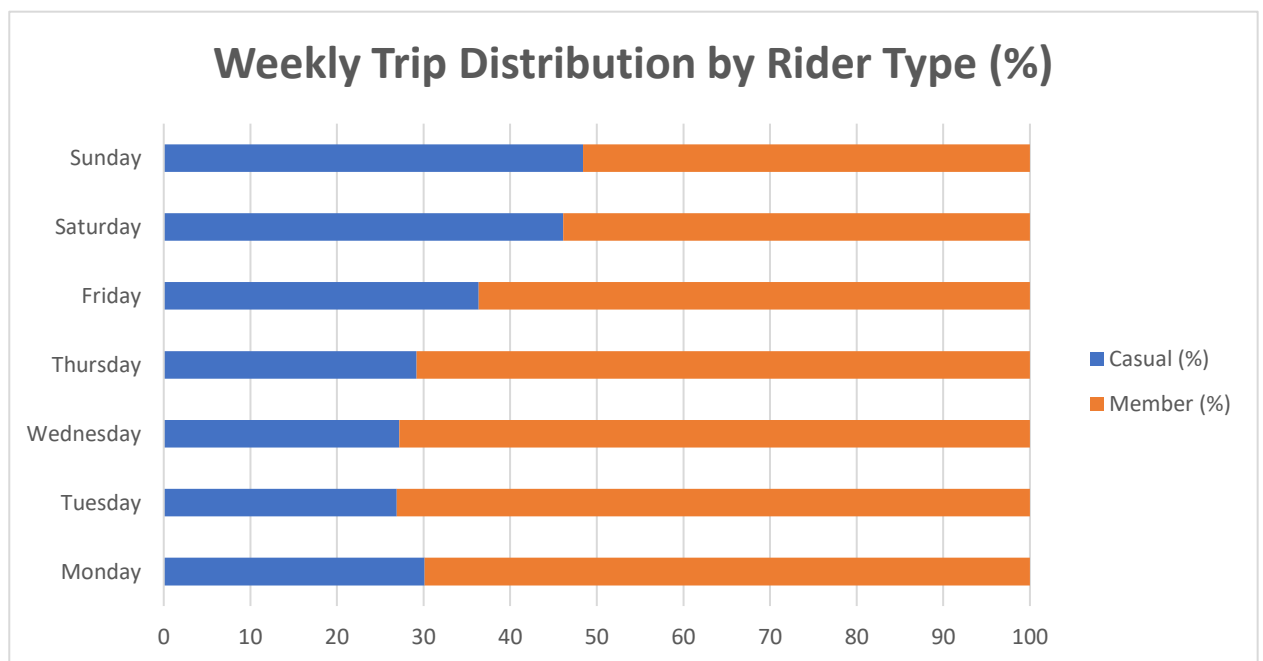


Key takeaway: Members generate most trips, but casual riders represent nearly 1.4M trips and therefore a substantial conversion opportunity.

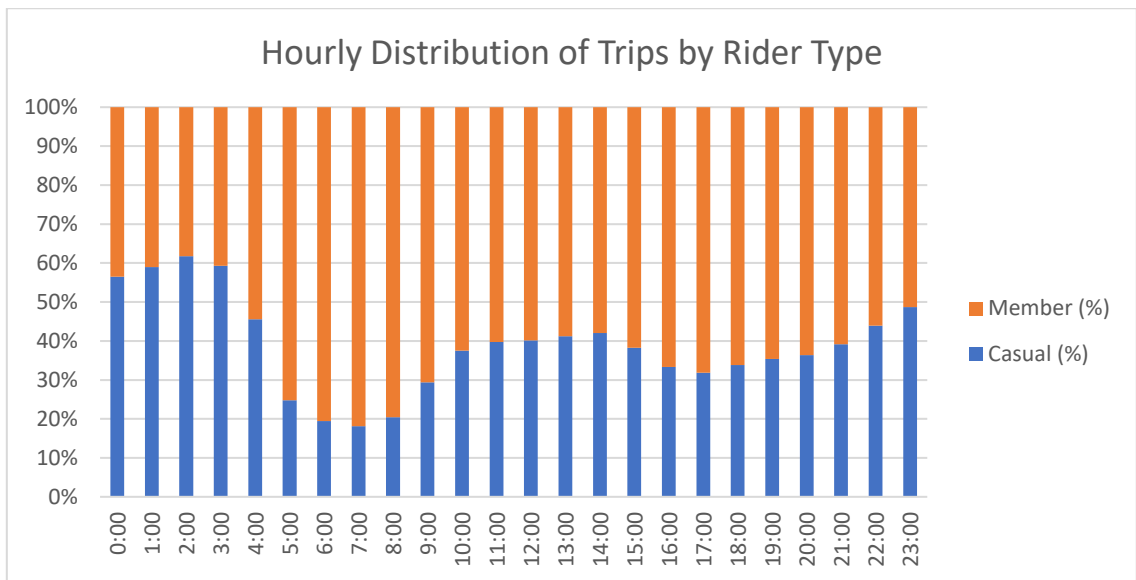
2. When are casual riders most active?



Key takeaway: Casual usage is strongest during warmer months.

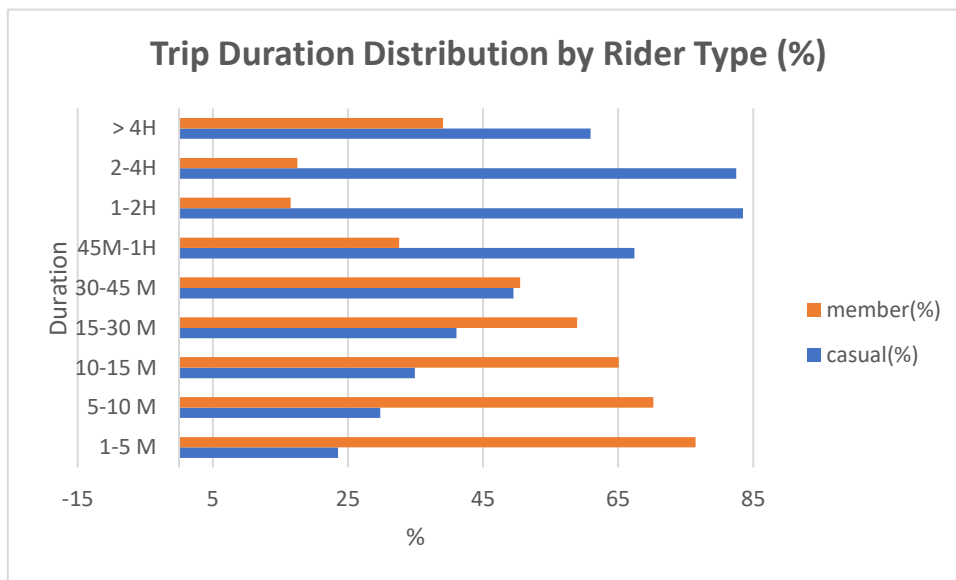


Key takeaway: Casual riders represent a much larger share of weekend trips.

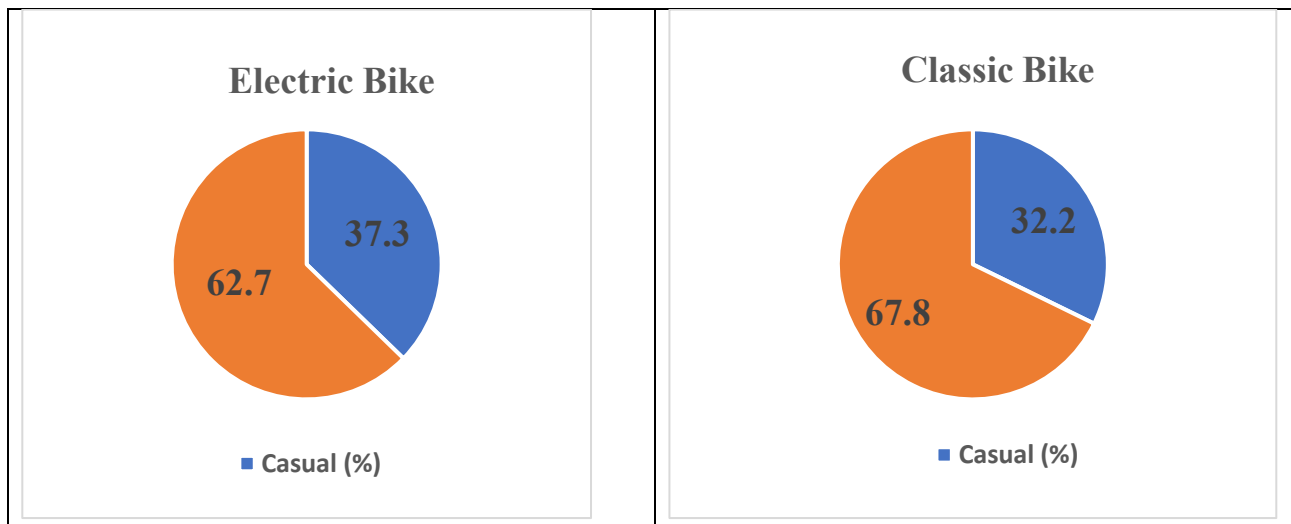


Key takeaway: Casual riders have their highest shares during overnight hours.

3. How do casual riders use the bikes?



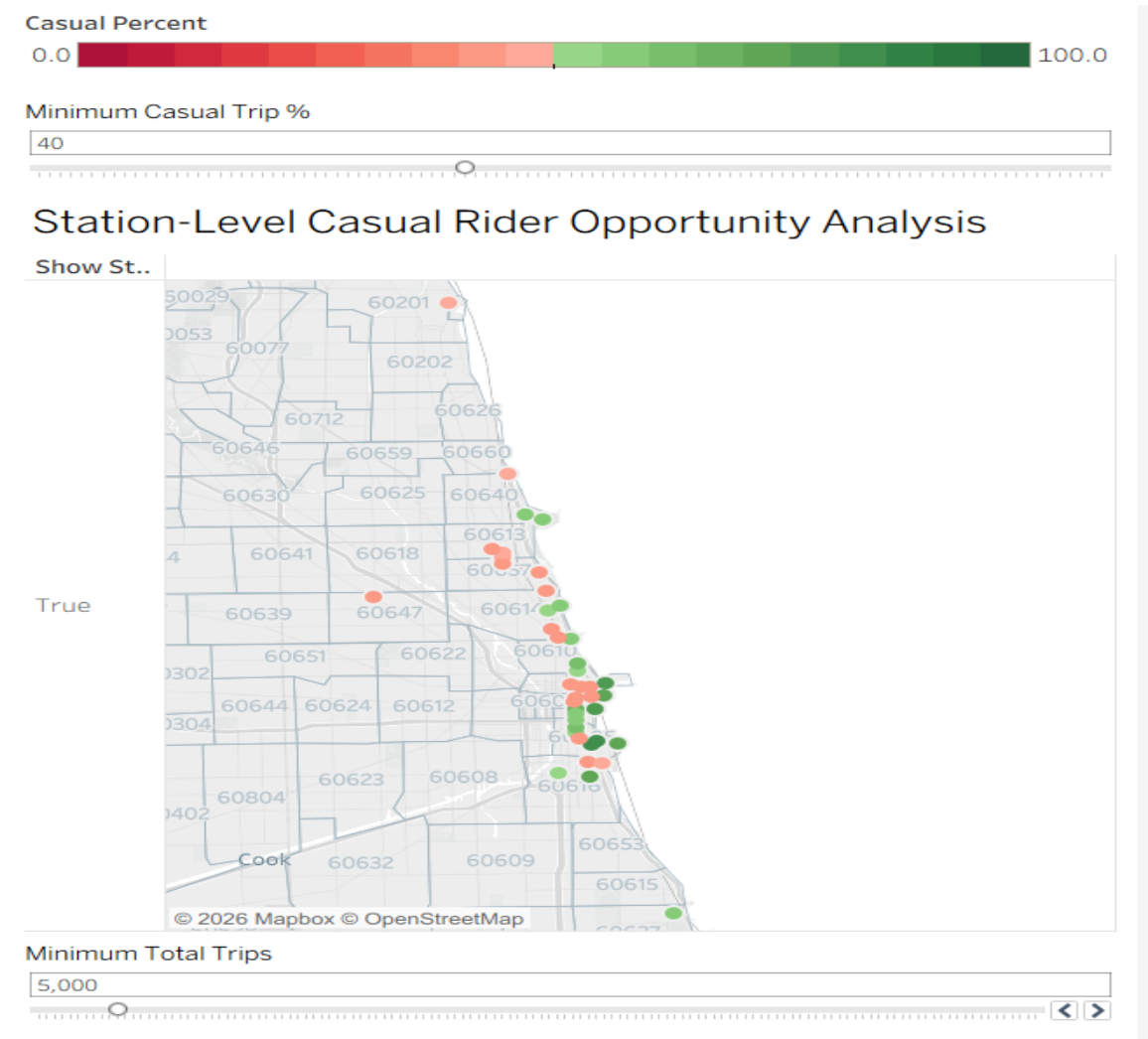
Key takeaway: Casual riders account for a much higher proportion of trips lasting more than 45 minutes.



Key takeaway: Electric bikes represent a larger share of casual trips than member trips.

4. Which stations should Cyclistic prioritize for membership-conversion campaigns?

Target criteria: Total trips $\geq 5,000$ and casual-trip proportion $\geq 40\%$.



- ✓ The target stations are concentrated primarily in central Chicago and areas along or near the Lake Michigan lakefront, including downtown, South Loop, River North, Lincoln Park, and Lakeview.

Act

Based on the analysis of casual and member rider behavior, Cyclistic should focus its membership-conversion strategy on **when casual riders are most active, where they are concentrated, and how their usage differs from members.**

Recommendation 1: Weekend membership campaigns

Finding: Casual riders represent a much higher proportion of weekend trips.

Action: Promote annual membership offers to casual riders during weekends.

Why: Weekend casual riders represent a large and identifiable audience for conversion.

Recommendation 2: Seasonal campaigns

Finding: Casual usage increases substantially during warmer months.

Action: Launch membership campaigns before and during the spring/summer riding season.

Why: Marketing efforts can reach casual riders when their engagement with Cyclistic is highest.

Recommendation 3: Target high-casual-use stations

Finding: Some high-volume stations have substantially higher casual-trip proportions.

Action: Prioritize these stations for membership messaging, signage, QR codes, or promotional campaigns.

Why: These locations provide direct access to riders who already actively use Cyclistic.

Recommendation 4: consider long-duration trips

Finding: Casual riders account for a much higher proportion of longer-duration trips.

Action: Test membership messaging that highlights the value and convenience of membership for frequent or longer rides.

Why: Longer-duration usage represents an identifiable behavioral segment that may be receptive to membership messaging.

Recommendation 5: Test rather than assume

Cyclistic should test different membership messages and offers rather than assuming that all casual riders respond to the same campaign. Campaign performance should be measured using conversion rate, membership purchases, and cost per converted member.

Limitations:

The analysis identifies behavioral associations between rider type, timing, duration, bike type, and station usage. It does not establish that these factors cause riders to remain casual or prevent membership conversion. The recommended campaigns should therefore be tested and evaluated using actual conversion results.

Closing Perspective

“I approach data with curiosity, attention to detail, and a focus on turning analysis into practical business insights.”